

Pandas Cheatsheet — Essential Functions (A4)

Data Loading & Inspection		Selecting & Filtering		Type Handling	
pd.read_csv("file.csv")	Load CSV	df['col']	Select column	df['id'].astype('int64')	Cast type
pd.read_json("data.json")	Load JSON	df[['a','b']]	Select multiple columns	df['id'].astype(str)	To string
pd.read_excel("file.xlsx")	Load Excel	df.iloc[0:5]	Select by position	pd.to_datetime(df['ts'])	To datetime
df.info()	Schema + nulls	df.loc[df['x']>0]	Select by label/condition	df['ts'].dt.date	Extract date
df.head(10)	Preview rows	df[df['age'] > 30]	Boolean filter	df['ts'].dt.year	Extract year
df.tail(5)	Last rows	df[(df['age']>18) & (df['country']=='US')]	Multi-condition	df['ts'].dt.month	Extract month
df.sample(5)	Random sample	df.query("country='US'")	SQL-style filter	df['ts'].dt.dayofweek	Day of week
df.shape	Rows/columns	df[df['age'].between(20,30)]	Range filter		
df.dtypes	Column types	df[df['id'].isin([1,2,3])]	Is-in filter		
df.columns	Column names				
df.describe()	Summary stats				

Cleaning & Transforming		Aggregation & Grouping		Joining & Combining	
df.dropna(subset=['id'])	Drop missing	df.groupby('cat')['sales'].sum()	Single col group	pd.merge(a, b, on='id', how='left')	Left join
df.dropna(how='all')	Drop all-null rows	df.groupby(['cat', 'region'])['sales'].sum()	Multi-col group	pd.merge(a, b, on='id', how='right')	Right join
df.dropna(inplace=True)	Drop missing in-place	df.groupby(['cat', 'region']).agg({...})	Multi-col + multi-agg	pd.merge(a, b, on='id', how='inner')	Inner join
df.fillna(0)	Fill missing	df.groupby('cat').mean()	Group mean	pd.merge(a, b, on='id', how='outer')	Outer join
df.fillna(0, inplace=True)	Fill in-place	df.groupby('id').agg({'a': 'sum', 'b': 'count'})	Multi-agg	pd.merge(a, b, how='cross')	Cross join
df.ffill()	Forward fill	df.groupby('cat').size()	Group size	df1.join(df2, how='left')	Index join
df.drop_duplicates(subset=['id'])	Deduplicate	df['id'].count()	Count	pd.concat([df1, df2])	Stack DataFrames
df.duplicated()	Check duplicates	df['user_id'].nunique()	Unique count	pd.concat([df1, df2], axis=1)	Concat columns
df.columns.difference(['id'])	Exclude columns	df['country'].value_counts()	Frequency		
df.reset_index(drop=True)	Reset index	df.sort_values('sales', ascending=False)	Sort		
df.reset_index(drop=True, inplace=True)	Reset index in-place	df.sort_values('sales', inplace=True)	Sort in-place		
df.rename(columns={'a': 'A'})	Rename	df.nlargest(5, 'sales')	Top N rows		
df.rename(columns={'a': 'A'}, inplace=True)	Rename in-place				
df.drop(columns=['col'])	Drop column				
df.drop(columns=['col'], inplace=True)	Drop column in-place				
df.replace({'A': 1})	Replace values				
df.assign(new_col=df['x']*2)	Add column				
np.where(df['x']>0, 1, 0)	Conditional column				

Advanced — Window & Cumulative		Advanced — Reshape & Strings	
df['ma7'] = df['sales'].rolling(7).mean()	Rolling mean	df.pivot_table(values='sales', index='cat', columns='region')	Pivot
df['lag'] = df['x'].shift(1)	Lag column	df.melt(id_vars=['id'])	Wide to long
df['r'] = df.groupby('cat')['sales'].rank()	Rank	df.explode('col')	Explode list col
df.groupby('cat').cumcount()	Cumcount per group	df['x2'] = df['x'].apply(lambda x: x*2)	Apply
df['x'].cumsum()	Cumulative sum	df['email'].str.lower()	String lower
df.groupby('cat')['x'].cumsum()	Cumsum per group	df['col'].str.contains('x')	String contains
df['x'].cummax()	Cumulative max	df['col'].str.split(',')	String split
df['x'].cummin()	Cumulative min	df.loc[df['age']<0, 'age'] = 0	Update rows
df['x'].pct_change()	Pct change		
df['x'].diff(1)	Diff (lag diff)		